

KASALITE - TECHNICAL DOCUMENTATION

THE PROBLEM

MamaCare AI is specifically developed for the AI for Reproductive Health in Africa Innovation Challenge (HASH) under Track I: Access to Comprehensive Early Pregnancy Loss Care.

Every year, women die from pregnancy complications that could have been prevented. According to the World Health Organization, approximately 287,000 women died from pregnancy-related causes in 2020 alone. That is nearly 800 women every single day. More than 94% of these deaths happened in low and middle-income countries, and sub-Saharan Africa accounts for about 70% of all maternal deaths worldwide.[1]

In West Africa, the maternal mortality ratio stands at approximately 542 deaths per 100,000 live births.[2] The gap is not explained by the complexity of the medical problems. The WHO estimates that more than 75% of all maternal deaths are preventable with timely care.[2] Women are not dying because we lack the knowledge to save them. They are dying because the care does not reach them in time.

Most maternal care is built around clinic visits. A woman attends her appointment, receives advice, and goes home. What happens between that visit and the next one is largely invisible to the health system.

This gap is where most deaths occur. Research shows that the majority of maternal deaths happen not during facility deliveries, but in the days and weeks before and after, when women are at home and out of contact with care providers. Data across multiple countries in sub-Saharan Africa shows that fewer than 43% of women received the recommended four or more antenatal care visits.[3] Postnatal care coverage is even lower, with fewer than 30% of women receiving a check within two days of delivery, which is the period of highest risk.[4]

The primary target users and beneficiaries of MamaCare AI are:

- **Pregnant and Post-Loss Mothers:** Specifically those residing in low-resource, rural, and peri-urban communities across Sub-Saharan Africa. These mothers typically rely on basic feature phones, have limited or no mobile data connectivity, and may communicate primarily in local dialects rather than English.
- **Community Health Workers (CHWs) & Midwives:** Frontline health workers who manage massive patient caseloads (often 200–400 patients per nurse) with paper-based systems. MamaCare AI acts as a triaging copilot, prioritizing their daily follow-up queue.
- **District Health Administrators:** Managers who require aggregate, real-time data on maternal outcomes and referral compliance to allocate scarce resources.

Maternal mortality in Sub-Saharan Africa remains a pressing humanitarian crisis, accounting for approximately 70% of all global maternal deaths. The maternal mortality ratio in West Africa stands at roughly 542 deaths per 100,000 live births, compared to fewer than 10 in high-income regions. Over 75% of these deaths are preventable with timely intervention.

Traditional digital health solutions rely heavily on smartphone applications or high-bandwidth internet connectivity. However, across rural Sub-Saharan Africa, fewer than one in three individuals have access to mobile internet, and a persistent 20% gender gap in mobile internet usage further disadvantages women. By contrast, basic voice-based GSM coverage reaches over 90% of the population. MamaCare AI leverages this existing low-tech telephonic infrastructure, communicating in native languages like Twi, Hausa, Swahili, and Amharic, bypassing barriers of digital illiteracy and lack of internet access.

SOLUTION OVERVIEW

MamaCare AI is a voice-based maternal care platform. It keeps women connected to health support throughout their pregnancy and recovery through regular, automated phone calls, while giving healthcare workers a clear, real-time picture of their patients so they can act quickly when someone needs help.

The platform was built around one central question: what does almost every woman in a low-resource setting already have access to? The answer is a basic phone. Not a smartphone. Not an internet connection. Just a phone that can make and receive calls. MamaCare AI works on that phone, in her language, without asking her to download anything or learn a new system. She just has to pick up when it calls. The platform integrates voice-based interaction with clinical risk detection, enabling automated check-ins and clinical decision support to manage both pregnancy and post-pregnancy loss cases efficiently.

When a woman is enrolled in MamaCare AI by a health worker, a community volunteer, or through a simple call herself, she begins receiving regular check-in calls. The timing of these calls is tied to her stage of pregnancy, so the questions she is asked are always relevant to where she is in her journey.

The calls feel like a conversation, not a questionnaire. The platform asks how she has been feeling, whether she has noticed any changes (such as bleeding, pain, or fever), and tracks her symptoms continuously. It listens to her answers and responds in a way that makes sense for her situation. If she mentions a headache, it asks a follow-up question. If she describes something that sounds like a warning sign, the system takes note.

When the platform detects something that may need attention, it acts. It may give her clear guidance on what to do, alert her health worker right away, or in urgent situations, contact the

nearest facility with her details and a summary of what she reported. This happens automatically, without her needing to know who to call or how to escalate the situation herself. Between these check-ins, the platform also supports her more broadly, reminding her when to take her medications, offering practical guidance on nutrition, and explaining clearly which symptoms mean she should go to a hospital without waiting.

When a pregnancy ends in loss, MamaCare AI does not go silent. The woman is moved into a dedicated care programme that is designed specifically for what she is going through, providing both physical monitoring and psychosocial support.

Healthcare workers access MamaCare AI through a simple web-based dashboard that works on any device, whether a desktop computer at the clinic, a tablet in the field, or a basic smartphone. The dashboard shows them all their patients, organised by how urgently each one needs attention.

User Workflow

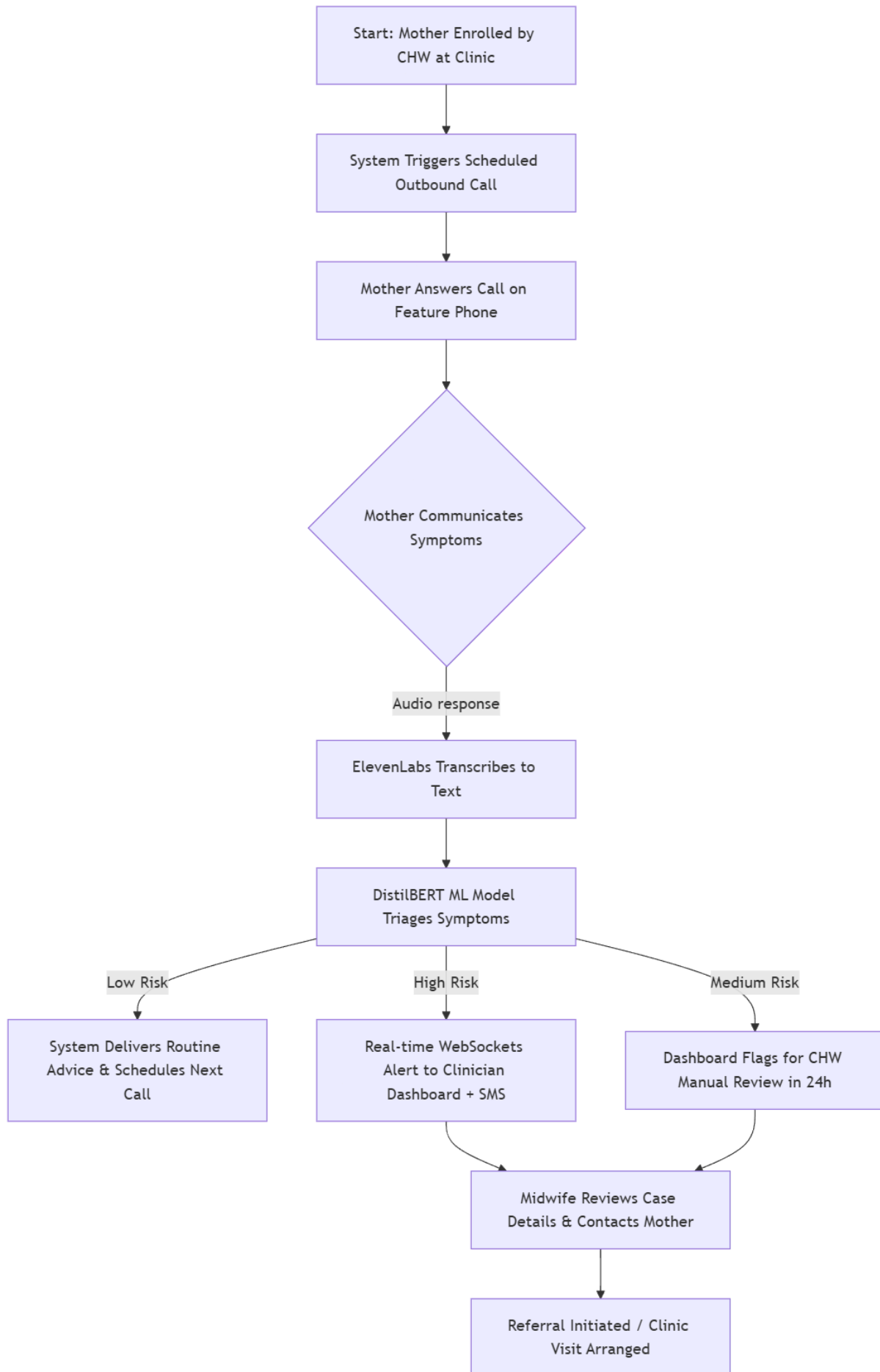
The patient-provider interaction follows a structured pathway to ensure prompt clinical response and triage:

1. **Enrollment:** The patient is enrolled in the clinic web interface by a midwife or community health worker, storing baseline details and language preference.
2. **Call Trigger:** The system schedules and initiates automated outbound calls based on the gestational calendar or recovery track.
3. **Voice Interaction:** The mother communicates her symptoms verbally. ElevenLabs processes the speech, transcribing it for analysis.
4. **Triage Assessment:** The custom-trained DistilBERT model classifies the transcript as Low, Medium, or High Risk based on Safe Motherhood guidelines.
5. **Escalation:** Medium and High-risk cases are immediately escalated to the Clinician Dashboard with real-time WebSocket alerts and SMS notifications to prompt immediate clinical review and direct outreach.

Expected Health System Benefits

- **Reduction in Maternal Mortality via Early Detection:** Continuous monitoring ensures that dangerous conditions like severe pre-eclampsia or postpartum hemorrhage are detected days before they reach a fatal crisis point, allowing for preventive clinical interventions.
- **Community Health Worker Optimization:** Triage automates the routine follow-up process, letting overstretched CHWs focus their limited hours on the high-risk cases that require human judgment and clinical care.

- **Elimination of Administrative & Paper Burden:** Automated record-keeping updates patient profiles after every call, eliminating manual registry logs and simplifying regional health reporting.
- **Continuity of Care for Pregnancy Loss:** Bridges the gap for post-loss patients who typically slide out of contact with the formal healthcare system, providing essential postpartum follow-up and clinical tracking.



TECHNICAL APPROACH

AI/ML Methods & Paradigm: **Guideline-Driven Few-Shot Fine-Tuning (GDF-FT)**

To ensure high classification accuracy without large clinical datasets, MamaCare AI utilizes Guideline-Driven Few-Shot Fine-Tuning (GDF-FT). Standard fine-tuning often struggles with edge cases or hallucinates diagnoses in low-resource languages. GDF-FT solves this by encoding clinical protocols directly into the weights of a multilingual encoder.

By fine-tuning distilbert-base-multilingual-cased using structured semantic anchors, the embedding space is forced to align clinical statements (e.g., English: "heavy vaginal bleeding") close to their corresponding triage weights.

2. Guideline Anchor Dataset

The training anchors are derived by us from the Ghana Health Service National Safe Motherhood Protocol (Revised Edition):

- HIGH RISK (Label 2)
- MEDIUM RISK (Label 1)
- LOW RISK (Label 0)

3. Data Processing & Fine-Tuning Pipeline

1. Oversampling: The high-fidelity clinical anchors (28 original rows) are oversampled 10x (280 rows) to balance them against the larger dataset.
2. Synthetic Integration: Combined with maternal urgency datasets (such as IDinsight's synthetic dataset). The training text is processed and tokenized using the multilingual DistilBERT tokenizer.
3. Training Parameters:
 - Base Checkpoint: distilbert-base-multilingual-cased (134M parameters)
 - Training Loop: Hugging Face Trainer for 3 epochs.
 - Learning Rate: 2e-5, with a weight decay of 0.01 and a batch size of 8.

4. Model Evaluation & Performance Metrics

The model was evaluated on a dedicated clinical test set:

- Overall Test Accuracy: 66.67%

Risk Class	Precision	Recall	F1-Score	Support
LOW (Label 0)	0.56	1.00	0.71	5
MEDIUM (Label 1)	1.00	0.20	0.33	5

Risk Class	Precision	Recall	F1-Score	Support
HIGH (Label 2)	0.80	0.80	0.80	5

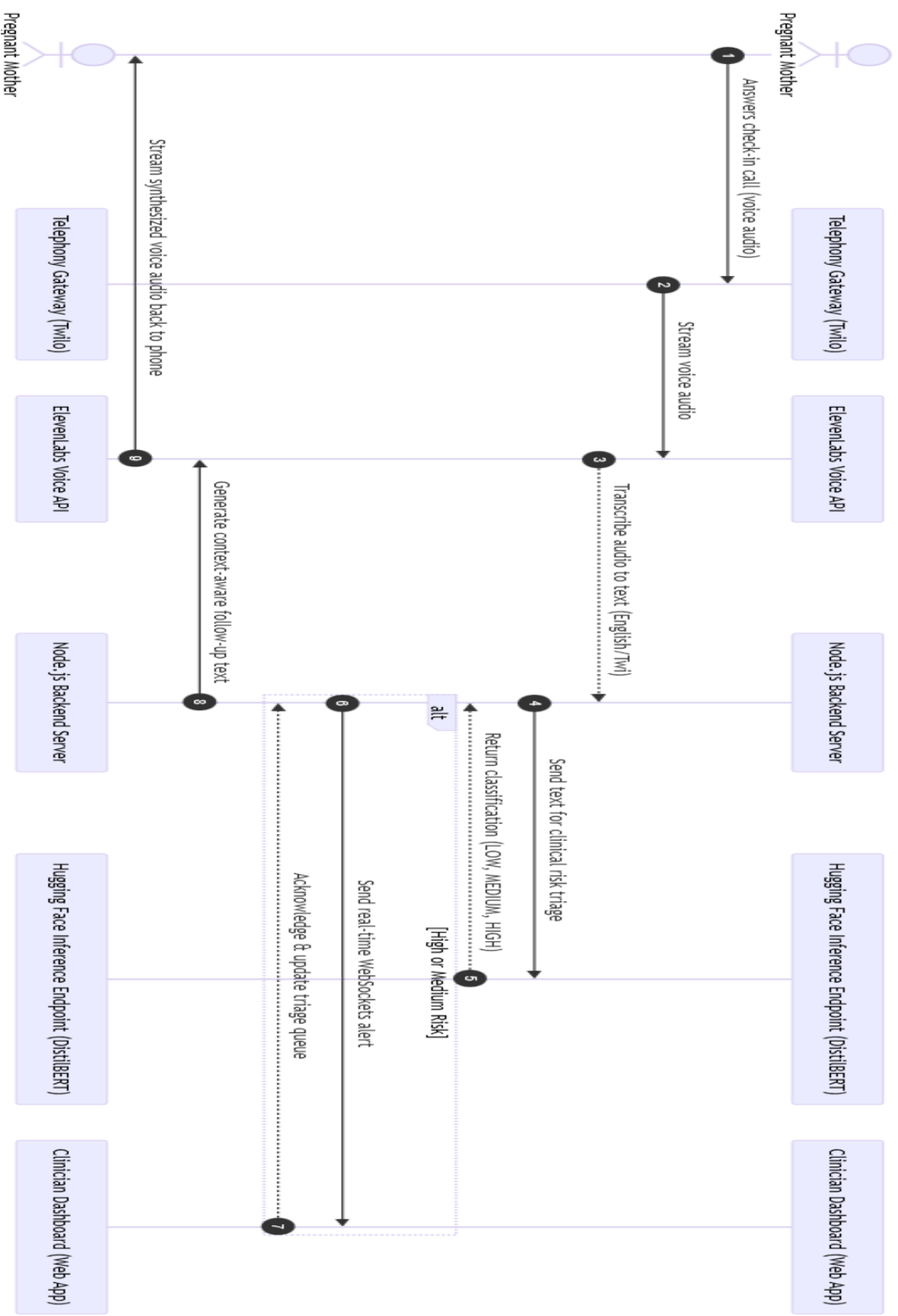
We acknowledge that more work needs to be done to effectively improve the model performance.

The model is hosted on the Hugging Face Hub at [sammydamz/mamacare-triage-model](https://huggingface.co/sammydamz/mamacare-triage-model).

5. Telephony Integration & ElevenLabs Architecture

MamaCare AI integrates telephony gateways with conversational speech systems:

1. **Call Processing:** Automated outbound calls are routed via Twilio to feature phones.
2. **Voice Generation & Transcription:** ElevenLabs is utilized to generate warm, empathetic, and culturally appropriate conversational voices .User voice responses are transcribed and routed to the classification backend.
3. **Inference & Alerts:** Transcripts are classified via the hosted Hugging Face Inference Endpoint. Risk detections immediately alert clinicians via WebSockets.



ETHICAL AND RESPONSIBLE AI CONSIDERATIONS

Data Privacy & Security

In many communities, reproductive status and experiences of pregnancy loss carry social sensitivity. Unauthorized disclosure can lead to social stigma or domestic harm.

- Protection: All data is encrypted in transit (TLS 1.3) and at rest (AES-256).
- Access Control: Role-Based Access Control (RBAC) ensures health workers can only see patients within their assigned facility.
- Consent: Consent is obtained verbally in the mother's local language during enrollment, explaining what data is monitored and that participation is entirely voluntary.
- Retention: Patient records are retained in compliance with local country guidelines and deleted upon request.

Bias & Fairness

Linguistic and demographic biases are common in global AI systems. MamaCare AI mitigates this by:

- Developing voice modules directly in local languages (Twi, Hausa, Swahili, Amharic) rather than relying on automated machine translation of English scripts.
- Using localized training anchors to ensure regional syntax is recognized.
- Training health workers to treat AI recommendations as clinical suggestions, leaving final diagnostic decisions to human expertise.

Transparency & Explainability

Rather than presenting black-box risk scores, the clinician dashboard displays plain-language symptom summaries derived from the rules in the National Safe Motherhood Protocol. For example, an alert will explicitly note: "Patient reported persistent headaches and facial swelling, indicating a high risk of Pre-Eclampsia."

SCALABILITY AND SUSTAINABILITY

Because it relies on standard GSM voice calls, MamaCare AI scales without requiring new physical infrastructure. Expanding to a new region requires only localized voice content adaptation and clinician dashboard onboarding, allowing the platform to grow rapidly.

The platform employs a business-to-government (B2G) SaaS model, contracting directly with national and district health authorities. This aligns with public health budgets and ensures the service remains free for patients, supplemented by development grants in its early stages.

MamaCare AI is designed to integrate into existing structures, such as Ghana's Community-Based Health Planning and Services (CHPS) framework. It acts as a tool to enhance the capacity of existing community health nurses, rather than creating a parallel, disconnected care delivery system.

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